

LIU ET AL. (2017)

Radiomics analysis for evaluation of pathological complete response to neoadjuvant chemoradiotherapy in locally advanced rectal cancer

Clinical Cancer Research

Citations: 1,200+

Impact Factor: 11.577

Category: Treatment Response

ABSTRACT SUMMARY

Key Contribution: pCR prediction in rectal cancer

Prediction of pathological complete response (pCR)

Pre-treatment radiomics signature development

Integration with clinical factors

Non-invasive treatment monitoring

Personalized treatment selection

METHODOLOGY ANALYSIS

□ STUDY DESIGN

• Dataset: 156 locally advanced rectal cancer patients

• Imaging: Pre-treatment T2-weighted MRI

• Features: 1,400+ radiomics features extracted

• Outcome: Pathological complete response (pCR)

□ pCR SIGNATURE DEVELOPMENT

• Validation: Cross-validation and independent testing

• Feature Selection: LASSO regression, CEA levels

• Signature Components: 15 key radiomics features

• Risk Stratification: High vs low response probability

• Validation: 5-fold cross-validation

• Clinical Integration: Combined radiomics + clinical factors

KEY RESULTS & FINDINGS

□ TREATMENT RESPONSE RESULTS

- Radiomics Only: AUC = 0.82 (strong baseline)
- Combined Model: AUC = 0.89 (significant improvement)
- Clinical Integration: 9% performance enhancement
- Risk Stratification: Clear response probability groups

□ CLINICAL IMPACT

- Validation: Robust cross-validation performance
- Non-invasive treatment response prediction
- Personalized treatment planning
- Reduced unnecessary surgeries
- Improved patient selection for organ preservation
- Clinical decision support for rectal cancer

INNOVATION & CLINICAL VALUE

□ INNOVATION

• First pCR prediction study in rectal cancer

• Pre-treatment response prediction

• Clinical-radiomics integration methodology

• Risk stratification for treatment planning

□ CLINICAL VALUE

• Non-invasive treatment monitoring approach

• Treatment response monitoring

• Patient selection for organ preservation

• Personalized treatment strategies

• Clinical trial stratification

• Quality of life improvement

LEGACY & FUTURE DIRECTIONS

□ □ LEGACY

• Foundation for treatment response prediction

• Clinical integration methodology

• Rectal cancer treatment planning

• Organ preservation strategies

□ FUTURE DIRECTIONS

• Clinical trial design

• Clinical implementation in rectal cancer

• Extension to other cancer types

• Real-time response monitoring

• Treatment optimization

• Clinical trial integration